



Where Mathematics Comes From: How the Embodied Mind Brings Mathematics into Being by
George Lakoff; Rafael Núñez

Review by: Jeffrey Nunemacher

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REVIEWS

Edited by **Gerald B. Folland**

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Where Mathematics Comes From: How the Embodied Mind Brings Mathematics into Being.
By George Lakoff and Rafael Núñez. Basic Books, New York, 2000, xvii + 493 pp. Hardcover:
ISBN 0-465-03770-4, \$30. Paperback: ISBN 0-465-03771-2, \$20.

Reviewed by **Jeffrey Nunemacher**

This provocative book is a first attempt to employ ideas from cognitive science and linguistics to understand the nature and origins of mathematics. Directed towards a reader who is somewhat knowledgeable about mathematics and interested in its foundations, the book is beautifully written and elegantly produced, with a dramatic picture of a bald head enveloped in arms on the dust jacket. The authors seek to explain mathematics as a system of metaphors that ultimately derive from basic neural processes. They analyze the structure of these metaphors and give several extended examples of their use to elucidate the basic number systems, the concept of infinity, and the mathematical theorem that $e^{\pi i} = -1$. Their results, if correct, have implications for both the philosophy of mathematics and mathematics education.

Neither of the authors is a trained mathematician, but they are ardent enthusiasts for mathematics, and they have consulted many well-known research mathematicians. The senior author, George Lakoff, is a prolific professor of linguistics at Berkeley who began his career as a student of Chomsky and was associated with the controversial generative semantics movement in the 1960s. He is an exponent of connectionism, the theory that behavior and thought should ultimately be explained in terms of physical connections between neurons. The junior author, Rafael Núñez, is a Swiss psychologist who has worked on cognition with applications to mathematics education. The present book is based on collaborative work carried out at Berkeley over the past several years.

My own reaction on reading the first few chapters was extremely positive. The approach is fresh and gives a plausible explanation for the origin of at least the most basic ideas in mathematics. I felt less satisfied, however, when the book tackled more advanced concepts, and I remain skeptical that “mathematical idea analysis” via conceptual metaphors provides a fruitful way to elucidate all mathematical ideas. I am particularly bothered by its inability to distinguish between correct and incorrect mathematics.

Lakoff and Núñez’s point of view is that all thought, in particular all mathematical thought, derives from basic neural connections. Thus there is nothing special about mathematics to distinguish it from other branches of human thought (and nothing special about language, either, which gets the authors into trouble with many linguists). Experimentation has shown that very young babies can distinguish immediately between collections of up to three or four objects. This innate ability to “subitize” (i.e., to recognize small collections) gives rise to an ability to perform addition and subtraction of very small natural numbers. From this foundation, the basic concepts of arithmetic are constructed by means of four “grounding metaphors”: object collection, from which the arithmetic of natural numbers is derived; object construction (that is,

breaking a whole down into pieces), which gives rise to the rationals; the measuring stick; and motion along a path, both of which lead to the reals.

A key principle in this analysis is that ideas originate in our innate ability to form conceptual metaphors. Here metaphor is meant not in the usual literary sense, but as a “grounded, inference-preserving cross-domain mapping” (page 6), which is typified by the idea of morphism in category theory. The authors create a formalism for expressing and applying such metaphors to particular situations, and they term this process “mathematical idea analysis.” Their approach is flexible and capable of yielding insights, and recent experiments in psychology and cognitive science provide empirical evidence for its validity. I found the analysis of the origins of arithmetic pretty convincing, and the metaphorical method does shed light on the nature of arithmetical ideas.

Another topic considered in this book is the origin and nature of the concept of infinity, which Núñez has studied in earlier work. The authors assert that all types of infinity in mathematics derive from the Basic Metaphor for Infinity (BMI), which takes “an iterative process which goes on and on” and completes it by adding a final state so that it can be regarded as a single entity. Thus from the potential infinity of $1, 2, 3, \dots$, one obtains the actual infinity consisting of the set of all natural numbers, which the authors regard as concluding with a final object ∞ . The cognitive origin of the BMI is taken to be the basic linguistic distinction between imperfective and perfective aspects of verbs, which describe uncompleted and completed actions. The book shows how this notion can be used to form various kinds of infinities such as the projective plane, transfinite numbers, and infinitesimals.

Some of these descriptions become rather complicated. For example, two levels of infinitesimal numbers are explored, partly to show that there is no single “true” conception of the real line. The first level contains the field of “granular numbers,” which is obtained from the reals by adjoining just one infinitesimal ∂ . At the second level are the much larger fields of nonstandard real numbers introduced by Abraham Robinson. I suspect that most readers will learn a bit of new mathematics in this section, and I wish that the authors had been a bit more explicit in giving references for further reading. (It is worth remarking that the bibliography for this book is quite extensive, but there is no attempt to provide citations for any particular constructions or ideas.)

I found the section on infinity less satisfactory than the one on arithmetic. The BMI is an interesting idea, but I am not convinced that it provides a natural or sufficient explanation for all uses of infinity. I don’t see why the basic model for infinite sets needs to contain a final element. Also, some of the applications of the BMI are so involved that the analyses do not enhance understanding. Presumably a cognitive description of transfinite numbers should not be more complicated than their mathematical construction. Finally, the inherent imprecision of a formal analysis of metaphor leaves me unsatisfied.

The book also contains cognitive analyses of various other mathematical concepts such as symbolic logic, geometry, and limits. Some of these treatments raise issues that we should consider as we teach the basics of our discipline. There was, for instance, a basic change in the conception of space that occurred in the process of laying axiomatic foundations for analysis and analytic geometry in the nineteenth century. Old-style space was the background in which geometric objects lived, whereas new-style space is the set of all points in a coordinate system. The authors rightly point out that such distinctions should be carefully addressed in the classroom, as they can be a source of difficulties for students who do not share the intuition of the instructor.

It seems to me that the authors occasionally confuse two related but rather different lines of study. The first is the investigation of the cognitive basis underlying human

mathematical thought, which is based on experiment and ultimately attempts to connect mathematical concepts with the neural structure of the brain. This is a proper domain for the methods of cognitive science, and when we learn enough about how the brain works, there should be a well-defined explanation. However, at our present level of knowledge, such an analysis of any complicated idea, mathematical or otherwise, seems beyond reach. The second line is “mathematical idea analysis,” which is the study of the heuristics that underlie mathematical ideas. (The last four chapters of the book are devoted to such an analysis of Euler’s equation $e^{\pi i} = -1$, with an explanation ultimately based on considering multiplication in the complex plane as a rotation.) Such a study is worth doing, but it has little to do with the (perhaps unknowable) cognitive origin of the ideas.

A serious problem with “mathematical idea analysis” is that heuristics cannot distinguish true results from false ones. Along the way to developing Euler’s equation, the authors attempt to arrive at the “meaning” of the number e . For this purpose they do a heuristic calculation (page 418) that leads from the condition $\lim_{q \rightarrow 0} b^q = \lim_{q \rightarrow 0} (1 + q)$ to the conclusion $b = \lim_{q \rightarrow 0} (1 + q)^{1/q}$. This sort of reasoning will disturb any mathematician, since it can just as easily be used to argue that from $\lim_{q \rightarrow 0} q^q = 1$ it follows that $\lim_{q \rightarrow 0} q = \lim_{q \rightarrow 0} 1^q$, i.e., that $0 = 1$.

The trouble with heuristics is that any explanation that is not totally precise is missing some of the subtlety of the concepts that it is trying to elucidate. A full examination of such subtleties is what constitutes a mathematical proof. The authors appear to believe that a proof is an idea-free formal argument, when in fact it is merely a careful explanation of why the asserted theorem is true. Thus “mathematical idea analysis,” properly understood, is not new; it is what properly written mathematics strives to do.

I imagine that the authors would respond to my complaint about the inability to distinguish between valid and invalid arguments by asserting that they are not doing mathematics but rather studying the concepts that underlie it; thus issues of validity do not come into play. If so, they provide no explanation for correctness, which is surely one of the most basic issues in mathematics, whatever one’s philosophical position on the ontology of mathematical concepts.

In the introduction to the book and in several of the later chapters, the authors explore the philosophical and pedagogical implications of their study. They take the position that, since mathematics consists of ideas that arise in a natural neural way and can be analyzed in the ways that they have illustrated, it has no special existence apart from being a human creation. This puts them in the anti-Platonic humanistic camp, today most closely associated with Imre Lakatos and Reuben Hersh (see, for example, Hersh [2]). They explicitly disavow the classical philosophies of formalism, logicism, Platonism, intuitionism, and social constructivism. (For a balanced treatment of many of these same philosophical issues by a logician, see Feferman’s readable collection of essays [1].) A practical moral that Lakoff and Núñez draw from their study is that the teaching of mathematics should be more focused on the exploration of the basic metaphors and heuristics that underlie mathematical cognition. The force of either of these conclusions is, of course, dependent on the persuasiveness of the argument for the connectionist view of mathematics offered in this book.

There are a few mathematical errors in the book, ranging from obvious slips such as an incorrect definition of the difference quotient (page 223) to more subtle misconceptions such as the assertion that an infinite chain of exponentiations of the infinite number $H = 1/\partial$ exists in the field of granular numbers (page 238). More disturbingly, no care is taken about operations on infinite series (page 413), and the impression is given that mathematical idea analysis frees one from the necessity to consider such technicalities.

In summary, this is an extremely attractive and ambitious book that attempts a cognitive explanation for the essence of mathematical concepts. It introduces and explores many intriguing ideas, but my opinion is that the authors have claimed substantially more than they have delivered.

REFERENCES

1. S. Feferman, *In the Light of Logic*, Oxford University Press, Oxford, 1998.
2. R. Hersh, *What Is Mathematics Really?*, Oxford University Press, Oxford, 1997.

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An Invitation to Algebraic Geometry. By K. Smith, L. Kahanpää, P. Kekäläinen, and W. Traves. Springer-Verlag, New York, 2000. ISBN 0-387-98980-3, \$32.95.

Reviewed by Mark Green

In educating mathematicians about a new area, it is worthwhile to distinguish between the courtship period and what comes after. Some books are written to make you fall in love with a subject; others are written to tell you what you need to know in order to live with it.

I vividly remember how I fell in love with algebraic geometry. I was a first-year graduate student at Princeton, heading for winter vacation. Thus far at Princeton, I had seen some lovely but quite abstract mathematics, but some good soul put a copy of Robert J. Walker's *Algebraic Curves* [10], a little Dover paperback, into my hands just before I left. When I reached Noether's theorem and its applications to the proof of Pascal's theorem and the group law for a plane cubic, something indefinable happened to me—admittedly, I was on a beach in the Virgin Islands, which may have had an effect—and somewhere in my brain the feeling that every author hopes to evoke was aroused: "I want to know more."

Algebraic geometry is not an easy subject either to teach or to learn. A list of worthwhile topics to know before undertaking the study of algebraic geometry could include commutative algebra, complex analysis, algebraic and differential topology, number theory, elliptic operator theory, K-theory, differential geometry, category theory, Lie groups and representation theory, several complex variables, and lattice polyhedra, with the recent essential addition of elementary particle theory. At the time, I had a decent understanding of perhaps two of these. Where to start?

I did not know it then, but the issue of how to help students get started learning algebraic geometry is a question that almost every practicing algebraic geometer would answer differently. Furthermore, at that time, books had not yet been written for many of the paths to algebraic geometry that are commonly trodden now. With the benefit of hindsight, I would distinguish four basic approaches:

1. *Schemes*. Known to its adherents as "doing it right the first time," the machinery of schemes was developed by Grothendieck in the 1960s, building on earlier work of